

Technical University of Denmark

Department of Health Technology

Sense-making of Raw Mobile Sensing Data

Mobile Sensing at UbiComp/ISWC 2025

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1 Background

UbiComp/ISWC is a well-known international conference where researchers, designers, developers, and practitioners gather to share and discuss new work in ubiquitous, pervasive, and wearable computing [1]. The 2025 edition took place at Aalto University in Espoo, Finland (October 1216, 2025), with the main conference running from October 1416 and pre-conference workshops on October 1213.

The **UbiLife** study collected mobile sensing data from around 82 participants, who carried smartphones running the CARP data collection app during the conference period (October 929, 2025). The dataset has over one million records across 15 different data streams, which covers passive sensors like step count, activity recognition, and location, as well as device data such as battery state, screen events, and heartbeat (keep alive) signals. It also includes environmental information such as weather and air quality, along with self-reported surveys on stress, sleep, wellbeing, personality, and session ratings. The main goal is to get a better understanding of how people move, behave and feel during a large academic conference, by looking at data collected from multiple sensor sources.

2 Data Exploration

2.1 Sensor Data

Table 1 summarises the 15 data streams with their record counts and participant coverage. Figure 1 provides a visual overview.

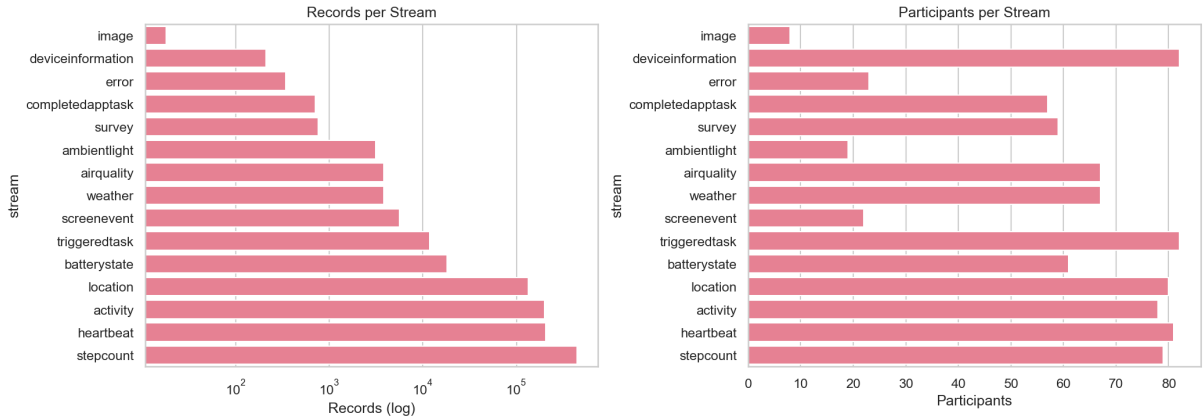


Figure 1: Overview of record counts and participant coverage across all 15 data streams.

Table 1: Record volumes and participant coverage per data stream.

Data Stream	Records	Participants
Step count	447 625	79
Heartbeat	204 811	81
Activity	199 336	78
Location	133 082	80
Battery state	18 119	61
Screen events	5 657	22
Weather	3 854	67
Air quality	3 837	67
Ambient light	3 115	19
Triggered tasks	11 795	82
Completed tasks	708	57
Survey	765	59
Device info	211	82
Error log	341	23
Image metadata	18	8

2.1.1 Temporal Coverage

Data collection spans October 9 to 29, with most activity happening during the main conference days (October 13 to 17). Figure 2 shows the heartbeat coverage heatmap: mean coverage during the conference was 33.4%, and night hours (00:00 to 06:00) had lower coverage across the board. All times in this report are local conference time (Europe/Helsinki, UTC+3), converted from the UTC timestamps recorded by CARP. Restricted to the four main conference days (October 13 to 16, Figure 3), 39 participants reach $\geq$10% coverage, with a mean of 37.6%. There is a large spread in how much data each participant contributed. Some recorded over 30 000 step count entries while others had fewer than 1 000 (Figure 4).

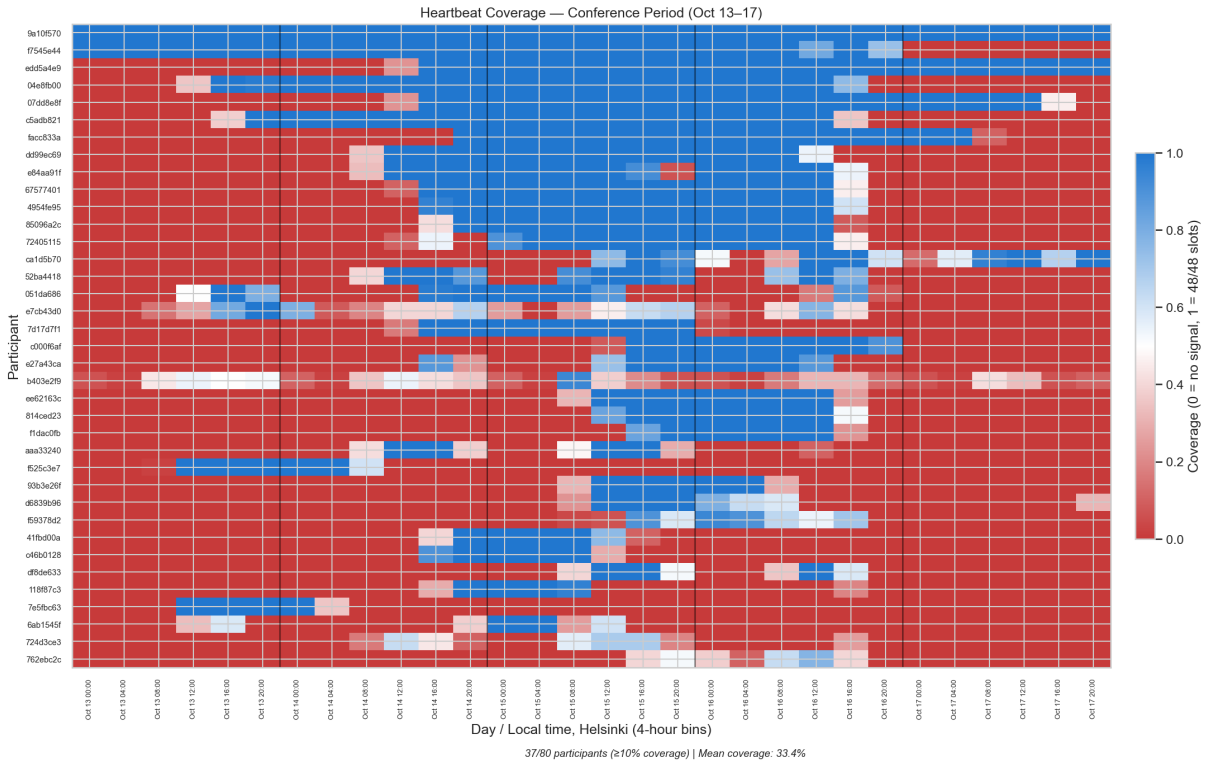


Figure 2: Heartbeat coverage heatmap (4-hour bins) for 37 participants with ≥10% coverage during October 13–17. Blue indicates high coverage; red indicates low coverage.

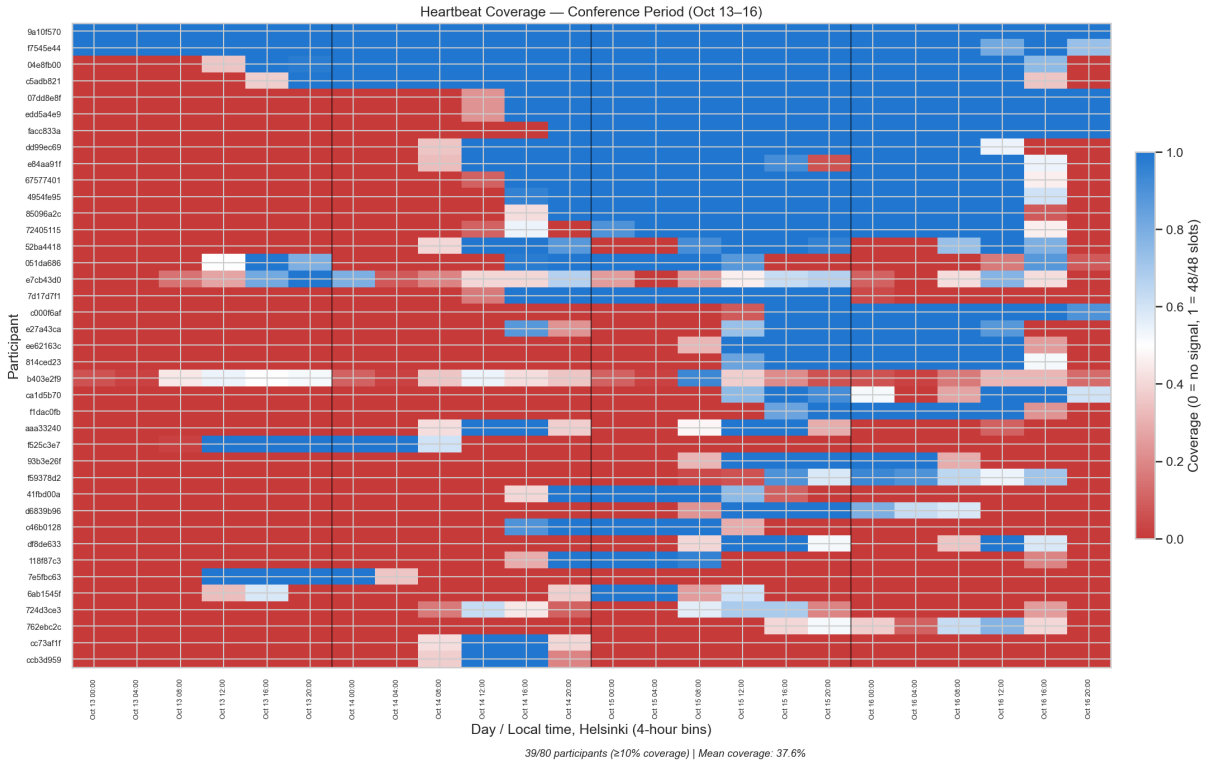


Figure 3: Heartbeat coverage heatmap (4-hour bins, local time) for 39 participants with ≥10% coverage during the main conference days, October 13–16.

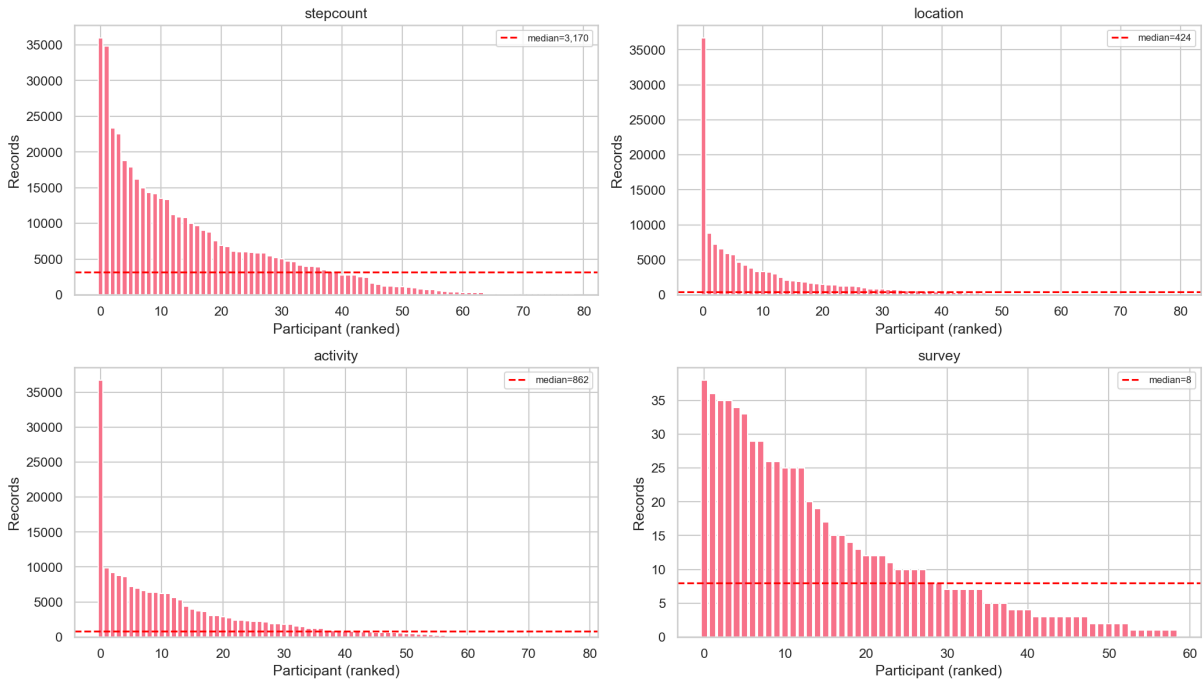


Figure 4: Distribution of records per participant across key sensor streams.

2.1.2 Activity Recognition

Five activity types were classified: STILL, WALKING, IN_VEHICLE, RUNNING, and ON_BICYCLE. STILL dominates between 00:00 and 06:00, while WALKING peaks during conference hours (09:00–18:00). Figure 5 shows the overall distribution and hourly patterns.

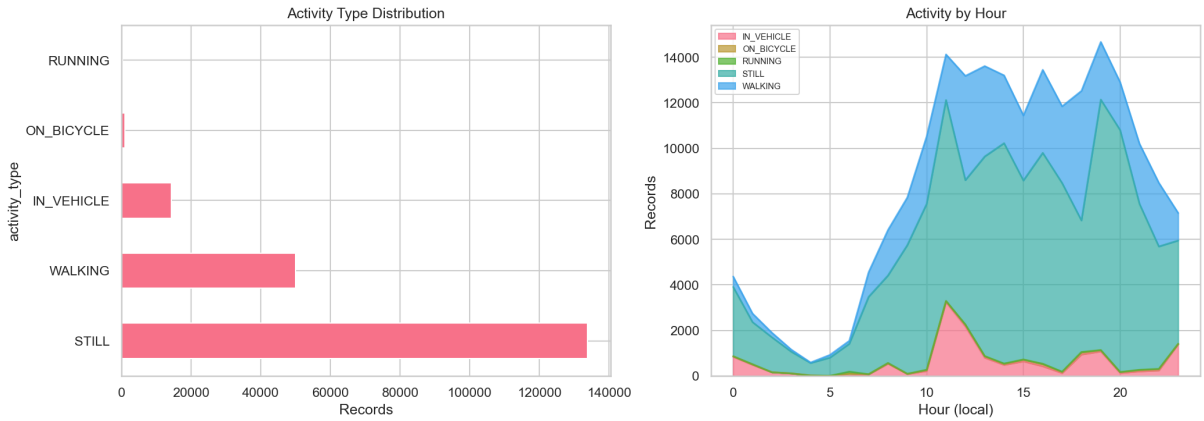


Figure 5: Activity type distribution and hourly activity patterns across all participants.

2.1.3 Device Diversity

Both iOS and Android platforms are represented, with manufacturers including Apple, Samsung, and Google (Figure 6).

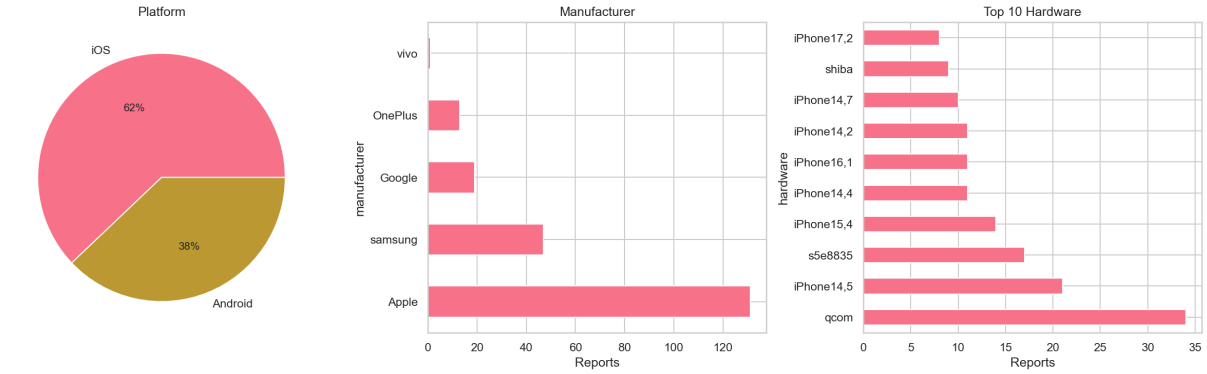


Figure 6: Device platform, manufacturer, and hardware breakdown.

2.2 Survey Data

Of 711 survey triggers (765 raw records minus 54 duplicate uploads), 561 were submitted (78.9% completion rate) and 150 were cancelled. Submissions peaked on October 14–16 (Figure 7). Table 2 summarises the main survey instruments.

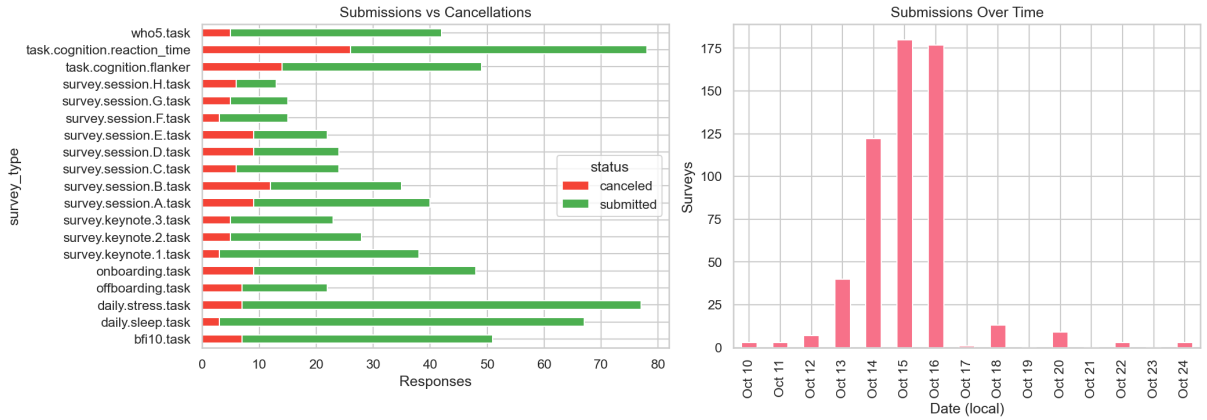


Figure 7: Survey submission status and temporal distribution of responses.

Table 2: Survey instruments and response counts.

Survey Type	Submitted	Participants
Daily stress	70	36
Daily sleep	64	36
Onboarding (demographics)	39	38
Cognition: reaction time	52	26
BFI-10 (Big Five)	44	42
Cognition: Flanker	35	23
WHO-5 (wellbeing)	37	37
Session ratings (A–H)	129	7–30
Keynote ratings (1–3)	76	18–35
Offboarding	15	15

2.2.1 Demographics

38 participants completed the onboarding survey (one response each), and all 38 gave valid age data (Figure 8). The median age was 30 years (mean 33.1, range 23 to 58), with both junior

and senior researchers represented. 57.9% were male ($n = 22$) and 42.1% female ($n = 16$).

In terms of academic roles, students made up 57.9% ($n = 22$), professors 21.1% ($n = 8$), postdocs 15.8% ($n = 6$), and researchers 5.3% ($n = 2$). More than half of the participants (57.9%, $n = 22$) were at UbiComp for the first time, while 23.7% had been to 4 to 6 previous editions and 7.9% had attended 7 or more.

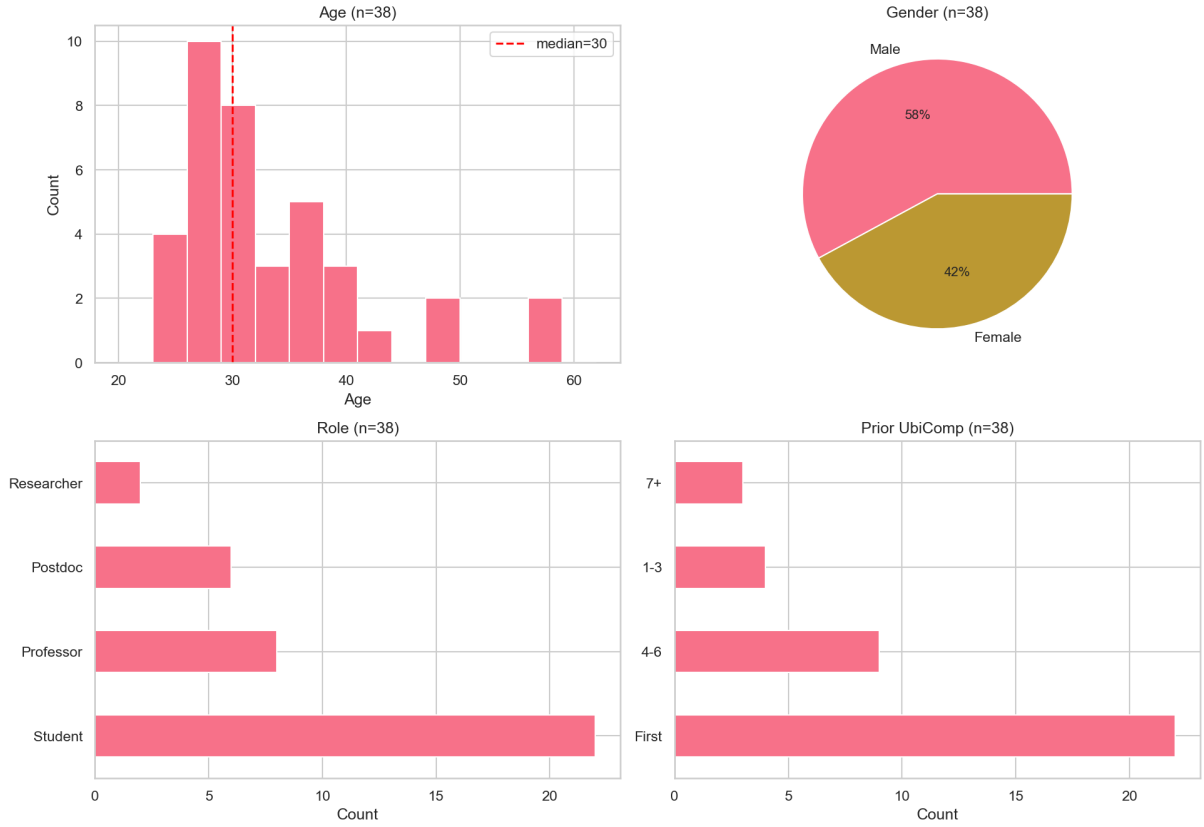


Figure 8: Participant demographics: age distribution, gender, academic role, and prior UbiComp experience.

2.2.2 Wellbeing: WHO-5

37 participants completed the WHO-5 Well-Being Index (Figure 9, left panel). Scores ranged from 28% to 84%, with a median of 60% and a mean of 57.4%. 10 participants (27.0%) scored below the 50% threshold that is used to screen for depression risk [2]. In the general population, this proportion is typically around 15 to 20%, so the higher rate here could be related to the travel, time pressure, and irregular schedules that come with attending a conference.

2.2.3 Personality: BFI-10

The BFI-10 (Big Five Inventory, 10-item short form) [3] was completed by 42 participants (Figure 9, right panel). Table 3 summarises the five personality dimensions (scale 0–4).

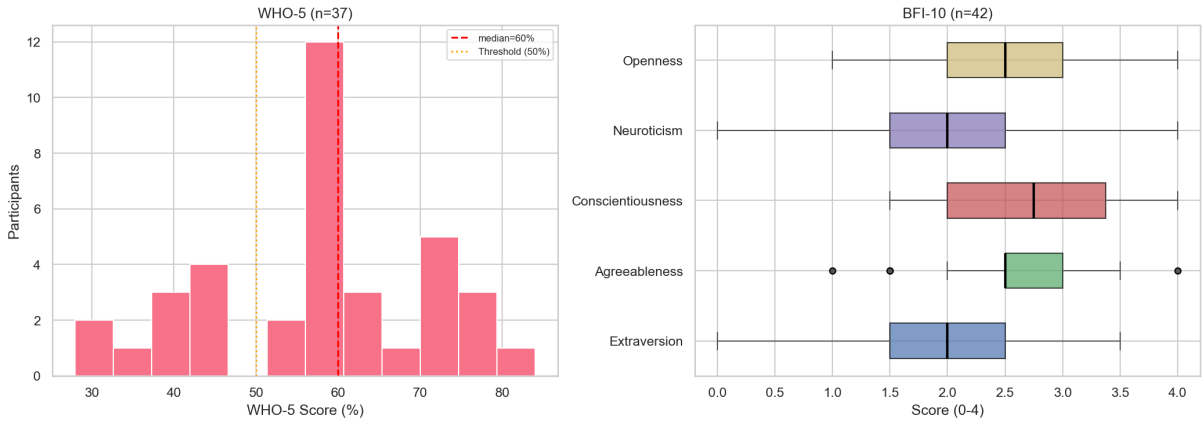


Figure 9: Left: WHO-5 wellbeing score distribution ($n = 37$); dashed line marks the 50% at-risk threshold. Right: BFI-10 personality dimension distributions ($n = 42$).

Table 3: BFI-10 personality dimension scores ($n = 42$, scale 0–4).

Dimension	**Mean**	**Std**
Conscientiousness	2.74	0.73
Agreeableness	2.68	0.62
Openness	2.42	0.71
Extraversion	2.07	0.76
Neuroticism	1.92	0.85

Conscientiousness had the highest mean score (2.74), which makes sense for an academic sample. Neuroticism was the lowest (mean 1.92) but also had the widest spread (std 0.85), so there was quite a bit of variation between participants. Extraversion scores were moderate (mean 2.07), meaning the group included both outgoing and more reserved people.

2.2.4 Daily Stress and Sleep

Daily stress was reported on a 0 to 10 scale, with 70 responses from 36 participants (Figure 10). The median stress level was 4.0 (mean 4.3, range 1 to 10), so stress was moderate on average.

For sleep quality ($n = 64$), most responses were positive: 51.6% said "Good", 26.6% "Neutral", 14.1% "Bad", 4.7% "Very Bad", and 3.1% "Amazing". Sleep duration paints a different picture though: 43.8% reported sleeping fewer than 6 hours, 51.6% slept 6 to 8 hours, and only 4.7% got more than 8 hours. So while people felt their sleep was okay, many of them were actually not getting enough hours, which is not unusual during a conference week.

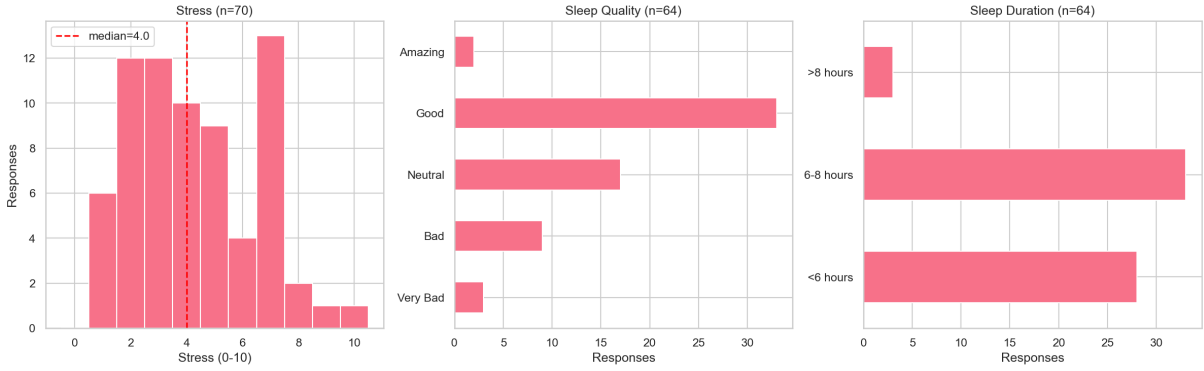


Figure 10: Daily self-report distributions: stress level (0–10), sleep quality, and sleep duration.

3 Data Preparation

3.1 Filtering to Conference Period

Timestamps were converted from UTC to local conference time (Europe/Helsinki) before filtering and any time-of-day analysis. All streams were filtered to the main conference week (October 13 to 16, 2025, local time). How much data was kept depends on the stream: screen events kept 99.1%, step count 87.9%, activity 83.3%, location 70.0%, heartbeat 51.9%, and ambient light 64.0%.

3.2 Step-Count Processing

Raw cumulative pedometer values were converted to per-interval step deltas:

1. Compute the first difference of the cumulative counter per participant.
2. Clip negative values (counter resets) to zero; fill NaN with zero.

3.3 Survey Parsing

The raw survey responses were stored as JSON columns with nested answer structures after parquet conversion. The protocol reuses the `onboarding.task` identifier for the offboarding survey, so offboarding responses were separated by their answer fields, and duplicate uploads (same participant, timestamp and survey) were removed. We wrote custom parsers to extract values and text labels for each survey type:

- **Onboarding**: age, gender, academic role, prior conference experience.
- **Daily stress**: single 0 to 10 Likert scale per day.
- **Daily sleep**: quality (categorical) and duration (categorical).
- **WHO-5**: five items scored 0 to 5; raw sum $\times$ 4 = percentage (0 to 100).
- **BFI-10**: ten items with reverse coding on selected dimensions; scores on five personality traits (0 to 4).
- **Event ratings**: session (A–H) and keynote (1–3) ratings with attendance confirmation.
- **Cognition**: reaction time (average reaction time, correct and wrong taps) and Flanker (score, swipes, mean congruent/incongruent response times); one row per test.

3.4 Processed Outputs

The processed data is about 22 MB across 17 Parquet files. Tables 4 and 5 list each file with its row count and columns. All sensor tables have common metadata columns (`participant_id`, `timestamp`, `sequenceId`, `studyDeploymentId`, `deviceRoleName`, `syncPoint.*`, `dataStream.*`). Below we only list the data columns specific to each stream. Note that sensor columns use the original CARP naming with a `measurement.data.` prefix (e.g. the full column name for `batteryLevel` is `measurement.data.batteryLevel`), while survey tables use cleaned snake_case names without any prefix.

Table 4: Processed sensor Parquet files (conference period, October 13–16).

File	Rows	Part.	Data Columns
stepcount	393 300	77	steps_cumulative, steps_delta
activity	166 047	77	type, confidence
location	93 209	78	latitude, longitude, altitude, speed, heading, accuracy, speedAccuracy
heartbeat	106 244	80	period, deviceType, deviceRoleName
weather	3 050	65	temperature, tempMin, tempMax, humidity, pressure, windSpeed, cloudiness, weatherMain, rainLastHour
airquality	3 033	65	airQualityIndex, airQualityLevel
batterystate	13 263	58	batteryLevel, batteryStatus
screenevent	5 606	22	screenEvent (ON / OFF / UNLOCKED)
ambientlight	1 993	19	meanLux, minLux, maxLux, stdLux

Table 5: Processed survey and cognition Parquet files (one table per survey type).

File	Rows	Part.	Columns
survey_onboarding	38	38	participant_id, timestamp, age, gender_value, gender_text, academic_value, academic_text, prior_value, prior_text, presenter_texts
survey_daily_stress	70	36	participant_id, date, timestamp, stress_level
survey_daily_sleep	64	36	participant_id, date, timestamp, sleep_quality_value, sleep_quality_text, sleep_hours_value, sleep_hours_text
survey_who5	37	37	participant_id, timestamp, q1, q2, q3, q4, q5, raw_sum, pct_score
survey_bfi10	42	42	participant_id, timestamp, extraversion, agreeableness, conscientiousness, neuroticism, openness
survey_event_ratings	205	38	participant_id, timestamp, attending_text, rating_value, rating_text, event_type, event_id
cognition_reaction_time	52	26	participant_id, date, timestamp, avg_reaction_time_ms, correct_taps, wrong_taps
cognition_flanker	35	23	participant_id, date, timestamp, score, time, right_swipes, wrong_swipes, mean_congruent_ms, mean_incongruent_ms, n_congruent, n_incongruent

3.5 Data Quality Notes

- **Heartbeat gaps**: mean coverage of 33.4% during the conference indicates frequent device disconnections.
- **Night-time blind spots**: all sensors show ∼50% lower coverage after midnight.
- **Per-participant variance**: step-count records range from <1 000 to >30 000 across participants.

References

[1] ACM, *UbiComp/ISWC 2025*, Aalto University, Espoo, Finland, October 12–16, 2025. https://www.ubicomp.org/ubicomp-iswc-2025/

[2] C. W. Topp, S. D. Østergaard, S. Søndergaard, and P. Bech, "The WHO-5 Well-Being Index: A systematic review of the literature," *Psychotherapy and Psychosomatics*, vol. 84, no. 3, pp. 167–176, 2015.

[3] B. Rammstedt and O. P. John, "Measuring personality in one minute or less: A 10-item short version of the Big Five Inventory in English and German," *Journal of Research in Personality*, vol. 41, no. 1, pp. 203–212, 2007.